Canvas3D: Object-Level 3D Lifting and Line-Art Composition for Painting Animation

Anonymous Authors
AIforEducation Project
aiforeducation@example.com

Abstract

Animating classical paintings into explorable videos requires both artistic fidelity and three-dimensional scene structure, yet existing methods offer only one or the other. Image-to-video diffusion models produce fluent motion but cannot maintain consistent 3D structure under viewpoint change, whereas 3D reconstruction pipelines capture geometry at the cost of painterly texture. We observe that the two objectives are hard to satisfy jointly because they favor conflicting representations, and propose **Canvas3D**, a framework that instead factors the problem into four sequentially composed sub-problems, each handled by a specialist. *PaintParse* decomposes the painting into foreground objects and a background layer, ordered by monocular depth; *MonoSplat* lifts each object into a metric-scaled 3D Gaussian representation and assembles a reusable object library; *SketchComp* treats hand-drawn line art as a spatial constraint, placing each 3D object by contour-based rigid registration and filling disocclusions with depth-conditioned inpainting; and *ArtiFlow* renders the composited scene along a camera trajectory and injects depth and edge cues as dual control signals into a video diffusion backbone, anchoring motion to explicit geometry. The decomposition supports object repositioning, multi-view exploration, and style-preserving animation—capabilities that per-frame conditioning cannot provide. On a benchmark of 50 classical paintings, Canvas3D outperforms image-to-video, reconstruction, and painting-specific baselines in style fidelity, perceptual quality, and temporal coherence, and user studies confirm a clear preference for its outputs.

1 Introduction

Classical paintings—oil on canvas, ink on silk, pigment on fresco—depict motion, narrative, and spatial depth that their static medium can only imply Lowenthal [2015]. Animating them as *explorable* videos, in which the viewpoint can move and painted subjects can be repositioned, has therefore become an active goal at the intersection of computer vision, generative modeling, and digital cultural heritage Dong [2024].

Progress on this goal, however, is split along two lines that do not meet. Image-to-video (I2V) diffusion models Blattmann *et al.* [2023]; Wan Team [2025]; Guo *et al.* [2024b]; Yang *et al.* [2025], together with conditioning mechanisms such as ControlNet Zhang *et al.* [2023a]; Wang *et al.* [2025a]; Peng *et al.* [2024], animate a single image into a short clip with plausible per-frame dynamics. Applied to paintings, they exhibit two shortcomings: they operate purely in 2D pixel space, with no notion of depth ordering, self-occlusion, or spatial layout, so they can neither synthesize novel viewpoints nor keep 3D structure consistent across frames; and temporal coherence degrades quickly beyond a few seconds, since latent interpolation has no geometric anchor for the motion trajectory. Reconstruction methods Mildenhall *et al.* [2020]; Kerbl *et al.* [2023]; Huang *et al.* [2024] provide the missing geometry, and feed-forward monocular variants VAST AI Research [2025]; Tang *et al.* [2024]; Charatan *et al.* [2024] can lift a single image to 3D, but they are optimized for photorealistic fidelity: applied to paintings, they replace brushstrokes with geometric artifacts, flatten impasto texture, and blur color boundaries under novel-view projection.

The underlying difficulty is that *3D structure and artistic style* favor conflicting representations, and neither model family achieves both. The conflict is most acute when the task demands **object-level manipulation**: moving a painted figure to a new compositional position while preserving its painterly identity and plausibly filling the disoccluded background.

Our approach. Rather than training one network to jointly optimize style and structure—an ill-posed objective whose gradients pull in opposite directions—we factor the problem into four sub-problems, each addressed by a specialist and connected to the next by a small, interpretable intermediate representation (Figure 1):

- **PaintParse** performs perceptual grouping: it decomposes the painting into foreground objects and a background layer with an art-domain-adapted

segmentation backbone, and orders the layers by monocular depth.

- **MonoSplat** handles geometry: it lifts each foreground region into a 3D Gaussian representation via feed-forward single-image prediction, rescaled to metric units by the PaintParse depth cue, and assembles an object-level library.
- **SketchComp** encodes compositional intent: it accepts hand-drawn line art specifying target positions, registers each 3D object to its sketched contour by a rigid transformation, and fills disocclusion gaps with depth-conditioned inpainting.
- **ArtiFlow** renders the composited scene into multi-view keyframes and drives a video diffusion backbone with dual control branches (depth and edge), so that object motion follows explicit 3D geometry rather than learned heuristics.

Each intermediate representation carries exactly what the next stage needs, which localizes errors instead of compounding them. The result is a system supporting object repositioning, multi-view exploration, and style-preserving animation—capabilities that no prior single-modality pipeline offers.

Contributions. Our contributions are:

1. A modular factorization of painting-to-video generation into four specialists—PaintParse, MonoSplat, SketchComp, and ArtiFlow—that communicate through small, interpretable intermediate representations; to our knowledge, this is the first end-to-end pipeline for 3D-aware, artistically controllable painting animation.
2. An object-level 3D lifting pipeline (MonoSplat) that couples art-domain-adapted segmentation with feed-forward monocular 3D prediction and depth-derived metric rescaling (Eq. 2), yielding a reusable library of metric-consistent 3D objects from a single painting.
3. A sketch-conditioned composition algorithm (SketchComp) whose contour-based rigid registration jointly optimizes region IoU and boundary Chamfer distance (Eq. 3), combined with a depth-conditioned inpainting module that fills disocclusion gaps (Eq. 4) while preserving background texture.
4. A geometry-grounded video synthesis workflow (ArtiFlow) whose dual control branches (Eq. 5) and first-frame latent anchoring bind temporal motion to explicit 3D structure with per-object control.
5. A comprehensive evaluation on a benchmark of 50 classical paintings, showing state-of-the-art style fidelity, perceptual quality, and temporal coherence, with user studies confirming preference for artistic authenticity, motion naturalness, and compositional-intent alignment.

2 Related Work

We review prior work along the four threads that motivate our decomposition and position Canvas3D at their intersection.

2.1 From Pixel-Level Stylization to Painting Animation

Early painting animation treats the input as a pixel grid to be stylized over time. Static style transfer Gatys *et al.* [2016]; Johnson *et al.* [2016]; Huang and Belongie [2017]; Park and Lee [2019]; Deng *et al.* [2022] established that artistic appearance can be re-injected into arbitrary content; its video extensions addressed temporal coherence via optical-flow regularization Ruder *et al.* [2016]; Chen *et al.* [2017]; Huang *et al.* [2017]; Wang *et al.* [2020]; Deng *et al.* [2021] and content deformation fields Ouyang *et al.* [2024]. A second lineage synthesizes the painting *process* itself, as stroke-level time-lapses Zhao *et al.* [2020]; Zou *et al.* [2021]; Liu *et al.* [2021]; Madono and Simo-Serra [2023], but reconstructs how a painting was made rather than how its subjects could move; a narrower thread animates painted faces Guo *et al.* [2024a]; Tan *et al.* [2024] only. Diffusion-era methods—inversion-based stylization Zhang *et al.* [2023b]; Chen *et al.* [2024], video editing Ceylan *et al.* [2023]; Chai *et al.* [2023]; Geyer *et al.* [2024]; Liang *et al.* [2024], and recent video stylizers Ye *et al.* [2025]; Li *et al.* [2026]; Jones *et al.* [2026]—preserve appearance under text-driven edits but provide no geometric handle on object pose. Domain-specific systems for Chinese ink painting Liang *et al.* [2021]; He *et al.* [2023]; Liu *et al.* [2026]; Zhang *et al.* [2025] follow the same pattern. Across all of these lines, appearance is preserved while spatial structure is not: no method offers object-level, 3D-grounded motion control.

2.2 3D Scene Representations and Single-Image Lifting

Scene representations evolved from implicit radiance fields Mildenhall *et al.* [2020] to explicit primitives: 3D Gaussian Splatting Kerbl *et al.* [2023] renders anisotropic Gaussians in real time, and 2D Gaussian Splatting Huang *et al.* [2024] improves surface accuracy. More relevant to us is the shift from per-scene optimization to feed-forward prediction: DreamGaussian Tang *et al.* [2024] generates 3D assets with generative priors, while pixelSplat Charatan *et al.* [2024], AnySplat Jiang *et al.* [2025], and TripoSplat VAST AI Research [2025] regress Gaussian parameters directly from one or few images. These predictors make it feasible to lift individual painting regions into 3D, but none addresses the composition of multiple lifted objects into a coherent, artistically controllable scene.

2.3 Sketch and Line Art as a Composition Interface

Sketches have long served as an interface for shape prototyping Olsen *et al.* [2009]. Recent systems use them

for generative ends: ShadowDraw Luo *et al.* [2025] synthesizes compositional shadow art, LineArt Wang *et al.* [2025b] transfers appearance onto design drawings, and region-aware simplification Nguyen *et al.* [2024] and volumetric animation toolkits Fox-Gieg [2024] demonstrate the expressive range of hand-drawn control. In diffusion pipelines, sketches typically act as appearance or boundary conditions, as in ControlNet-style edge injection Zhang *et al.* [2023a]. Canvas3D instead treats line art as a *spatial constraint*: the sketch is a target silhouette to which a 3D object is geometrically registered, resolving the which-object-where problem in scene assembly.

2.4 Controllable Image-to-Video Generation

Diffusion transformers Peebles and Xie [2023] and video diffusion models Ho *et al.* [2022a,b] underpin a family of scalable I2V backbones Blattmann *et al.* [2023]; Wan Team [2025]; Guo *et al.* [2024b]; Yang *et al.* [2025]. Controllability is usually pursued by auxiliary conditioning: ControlNet Zhang *et al.* [2023a] injects edge, depth, and pose maps, with video extensions Wang *et al.* [2025a] and efficient variants Peng *et al.* [2024]; cinematic shot design Xing *et al.* [2025] and clipart animation Wu *et al.* [2025] achieve compelling results without explicit 3D supervision. All of these condition *per frame*: they cannot reason about object identity, depth ordering, or occlusion across frames, and drift under viewpoint change. Canvas3D grounds its control signals in an explicit, object-level 3D scene that is composited and rendered before synthesis.

2.5 Painting-Specific Animation Pipelines

A small body of work targets paintings directly. Domain-adapted detectors Chen *et al.* [2025] and art datasets Saleh and Elgammal [2016] show that natural-image backbones Jocher *et al.* [2023]; Wang *et al.* [2024b,a]; Tian *et al.* [2025] and segmentation foundation models Kirillov *et al.* [2023]; Ravi *et al.* [2024] require art-specific fine-tuning, due to brushstroke texture and intentional boundary blending. Closest to our setting, "Printing the Underdetermined" Ming *et al.* [2026] translates paintings to 3D via video generative models but processes the image as a whole, and OilPaintingRevive Anonymous [2026] combines 2DGS with video refinement yet relies on per-scene skeleton binding that limits compositional flexibility. Canvas3D differs on four counts: object-level rather than image-level manipulation, line art as the composition interface, depth-aware inpainting for disocclusion, and multi-modal geometric conditioning for video synthesis.

3 Method

Design philosophy. A single painting-animation system must simultaneously satisfy four *conflicting* objectives: perceptual grouping of artistically rendered content, geometric reconstruction of objects that have no real 3D extent, user-specifiable composition, and temporally coherent stylized rendering. Jointly optimizing a single network for all four is ill-posed because the gradients of style fidelity and 3D structure pull in opposite directions. Our central design principle is therefore *modular separation of concerns with explicit handshakes*: each objective is addressed by an independently optimized specialist, and the four specialists communicate only through well-defined, interpretable intermediate representations—masks with depth-ordered layers, a metric-scaled object-level 3D library, a composited 3D scene with disocclusion-filled background, and finally a multi-view keyframe stream with paired geometry and edge cues. This decomposition converts a conflicting joint problem into four well-posed sub-problems, while the handshakes carry exactly the information each downstream stage needs and no more, preventing error compounding.

Problem statement. Given a painting scan $P \in \mathbb{R}^{H\times W\times 3}$ and a set of hand-drawn line-art sketches $L = \{\ell_1, \dots, \ell_K\}$ specifying target object positions, Canvas3D produces an animated video V that (i) preserves the artistic style of P, (ii) supports viewpoint exploration of a 3D-composited scene, and (iii) places each foreground object at its sketched location with geometrically grounded, temporally coherent motion. The framework comprises four modules (Figure 1): **PaintParse** (perceptual grouping and layer decomposition), **MonoSplat** (per-object 3D lifting and library assembly), **SketchComp** (line-art-guided scene composition with depth-aware inpainting), and **ArtiFlow** (geometry-grounded video synthesis).

Notation. Let $\Omega = \{1, \dots, H\} \times \{1, \dots, W\}$ denote the pixel grid of the painting. For the i-th foreground object ($i \in \{1, \dots, K\}$), $M_i \in \{0,1\}^{H\times W}$ is its binary mask and $\Omega_i = \{p \in \Omega \mid M_i(p) = 1\}$ the associated pixel set; the residual background mask is $M_{\text{bg}} \in \{0,1\}^{H\times W}$. The (metric-aligned) monocular depth map is $D_{\text{mono}} : \Omega \to \mathbb{R}_+$; each object is summarized by its *median* depth $d_i = \text{median}\{D_{\text{mono}}(p) \mid p \in \Omega_i\}$, used for metric recovery, together with a robust depth range $[d_i^{\text{near}}, d_i^{\text{far}}]$ (the 10th and 90th percentiles over Ω_i), used only for layer ordering. The 3D representation of object i is $\mathcal{G}_i$ and the assembled library is $\{\mathcal{G}_1, \dots, \mathcal{G}_K\}$; the composited scene and its depth map are $\mathcal{S}^*$ and $\mathcal{D}^*$; Ω_{gap} is the disocclusion region; and $\{I_t, D_t, E_t\}_{t=1}^T$ are the multi-view RGB, depth, and edge keyframes rendered along camera poses $\{\mathbf{T}_t\}_{t=1}^T$. The focal length in pixels is f. Throughout, $\pi(\cdot)$ denotes the perspective projection onto the image plane and $\text{IoU}(\cdot,\cdot)$ the intersection-over-union between two binary regions.

3.1 PaintParse: Art-Domain Perceptual Grouping and Layer Ordering

Design rationale. The downstream 3D and composition stages require two distinct outputs that no single head of an off-the-shelf detector provides: *(i)* foreground masks whose boundaries respect artistic intent rather

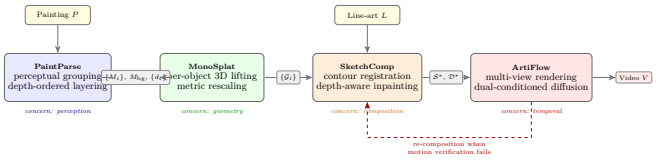


Figure 1: **Canvas3D overview.** The pipeline factors painting animation into four independently optimized specialists—*PaintParse* (perceptual grouping and depth-ordered layering), *MonoSplat* (per-object 3D lifting into a reusable, metric-scaled library), *SketchComp* (line-art-guided scene composition with depth-aware inpainting), and *ArtiFlow* (geometry-grounded video synthesis). Adjacent specialists communicate exclusively through small, interpretable *handshake representations* (gray tokens): depth-ordered masks, a metric 3D object library, and the composited scene with its depth map. The dashed loop re-enters SketchComp when ArtiFlow's per-object motion verification fails.

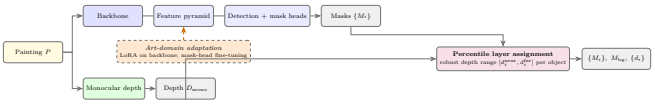


Figure 2: **PaintParse.** A segmentation branch (top) and a monocular depth branch (bottom) run in parallel on the input painting. Art-domain adaptation (dashed) injects painting-specific statistics into the backbone and sharpens the mask head. A percentile-based layer assignment fuses both outputs into depth-ordered layers, treating each object as an atomic unit.

than photographic edges, and *(ii)* a depth-consistent layer ordering that disambiguates which object lies in front of which. PaintParse therefore couples a domain-adapted instance segmentation backbone with a parallel monocular depth estimator and fuses their outputs through a percentile-based layer assignment. The key design choice is *decoupling semantic grouping from depth ordering*: the segmentation network is responsible only for "what is an object," while the depth estimator is responsible only for "how far away it is," and the layer-assignment logic simply reads both off. This avoids asking either network to solve a problem it was not trained for.

Instance segmentation with art-domain adaptation. We adopt a one-stage instance segmentation backbone pre-trained on natural-image data Jocher *et al.* [2023]. Direct transfer to paintings yields poor mask quality because brushstroke textures produce spurious edge responses and depth-ambiguous compositions confound object–background separation. We mitigate this through a two-stage adaptation protocol: a *low-rank adaptation* Hu *et al.* [2022] of the detection backbone, which injects art-domain statistics while constraining the parameter drift that would cause catastrophic forgetting, followed by *full fine-tuning* of the mask-prediction head to align boundary prediction with painterly edges. Given input P, the adapted model outputs K binary masks $\{M_i\}_{i=1}^{K}$, ordered such that $\Omega_i \cap \Omega_j = \varnothing$ for $i \neq j$ (overlapping detections are suppressed by non-maximum suppression and mask deduplication). The residual background mask is the complement of the foreground union:

$$M_{\text{bg}}(p) \;=\; 1 \;-\; \max_{i\in\{1,\dots,K\}} M_i(p), \qquad \forall p \in \Omega. \quad (1)$$

Monocular depth and layer assignment. In parallel, a monocular depth estimator Yang *et al.* [2024] produces a relative depth map $D_{\text{mono}} : \Omega \to \mathbb{R}_+$ (nearness-coded: smaller values indicate closer content). For each detected object i we summarize its depth by the 10th and 90th percentiles of $\{D_{\text{mono}}(p) \mid p \in \Omega_i\}$, yielding the robust range $[d_i^{\text{near}}, d_i^{\text{far}}]$ that is insensitive to outlier pixels at mask boundaries, and by the median $d_i = \text{median}\{D_{\text{mono}}(p) \mid p \in \Omega_i\}$, which the MonoSplat stage consumes for metric recovery. The layer index of each object is determined by the depth percentile of its mask centroid, i.e., object i is assigned a *lower* (front) layer than object j whenever the centroid depth of i is smaller. This percentile-based assignment is preferable to a per-pixel depth sort because it treats each object as an atomic unit, preserving its identity under the rigid transformations that subsequent composition applies.

3.2 MonoSplat: Per-Object 3D Lifting and Library Assembly

Design rationale. A whole-scene 3D reconstruction would entangle object geometry with background texture, making independent repositioning impossible. We therefore lift each foreground object *separately* into a 3D representation and assemble a *reusable library*, indexed by semantic label and original painting coordinates. Two design choices follow from this principle. First, we use a *feed-forward single-image 3D predictor* rather than per-scene optimization, because per-object optimization would be prohibitively expensive and would overfit to a single viewpoint. Second, because feed-forward predictors operate in a normalized canonical space without metric scale, we *recover metric dimensions from the PaintParse depth cue*, so that objects from different paintings can later be composed in a common metric frame.

Feed-forward lifting. For each foreground object i, we crop the painting to its mask, $P_i = P \odot M_i$, and feed the crop into a feed-forward single-image 3D predictor VAST AI Research [2025] that regresses a set of N_i anisotropic 3D Gaussians directly from one image. Each Gaussian j is parameterized by a center $\boldsymbol{\mu}_j \in \mathbb{R}^3$, scales $\mathbf{s}_j \in \mathbb{R}^3_+$, rotation unit quaternion $\mathbf{q}_j \in \mathbb{S}^3 \subset \mathbb{R}^4$, opacity $\alpha_j \in [0,1]$, and view-dependent color coefficients, and the object is rendered by α-composited Gaussian splatting. The predictor's generative prior, trained on diverse 3D assets, generalizes to the missing side and back of objects that are not visible in the painting—a critical capability for the rotations that downstream composition imposes.

Metric rescaling. The predictor outputs geometry in a normalized canonical space, i.e., it estimates shape $\widehat{\mathcal{G}}_i$ up to an unknown similarity transform. To place objects

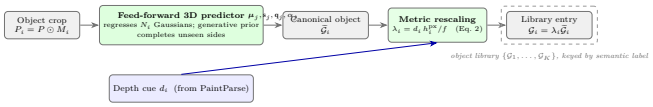


Figure 3: **MonoSplat.** Each foreground crop is lifted in a single feed-forward pass into a canonical-space Gaussian representation, whose generative prior completes the object sides absent from the painting. The PaintParse depth cue d_i restores metric scale (Eq. 2), placing every object in a common metric frame so that the resulting library entries can be composed across paintings.

from different paintings (and different object classes) in a single metric frame, we recover the scale factor from the pinhole imaging relation. Under the standard thin-lens approximation, the physical height h_i of an object, its image-plane extent h_i^{px} in pixels, its depth d_i, and the focal length f (in pixels) satisfy $h_i = d_i\, h_i^{\mathrm{px}}/f$. We therefore rescale the canonical geometry by

$$\lambda_i \;=\; \frac{d_i \cdot h_i^{\mathrm{px}}}{f \cdot h_i^{\mathrm{can}}}, \tag{2}$$

where d_i is the median depth of object i (from PaintParse), h_i^{px} is its mask height in the painting, and h_i^{can} is the height of its bounding volume in the predictor's canonical output space; the ratio $h_i^{\mathrm{px}}/h_i^{\mathrm{can}}$ converts the canonical unit to pixels before the depth relation maps it to metric units. Each Gaussian center and scale is rescaled as $\boldsymbol{\mu}_j \mapsto \lambda_i \boldsymbol{\mu}_j$, $\mathbf{s}_j \mapsto \lambda_i \mathbf{s}_j$, producing a metric-consistent 3D object $\mathcal{G}_i = \lambda_i \widehat{\mathcal{G}}_i$. The resulting library $\{\mathcal{G}_1, \ldots, \mathcal{G}_K\}$ is keyed by semantic label and original painting coordinates, enabling selective retrieval and cross-painting composition in future work.

3.3 SketchComp: Line-Art-Guided 3D Scene Composition

Design rationale. The user's compositional intent—which object goes where—is most naturally expressed as a sketch, but existing diffusion-based sketch control treats sketches as appearance or boundary cues rather than spatial constraints. SketchComp inverts this convention: a sketch *is* the target, and a 3D object is *geometrically registered* to it. Two sub-problems must then be solved: *(i)* finding the rigid transformation that aligns each 3D object's rendered silhouette with its sketched contour, and *(ii)* filling the disocclusion gaps that open up when an object moves away from its original position, while preserving the background's artistic texture. We solve the first with a silhouette-overlap objective that combines region IoU with boundary Chamfer distance, and the second with a depth-conditioned inpainting module whose geometric conditioning prevents the texture-smear artifact that naive inpainting produces on paintings.

Contour-based rigid registration. Each sketch ℓ_i is converted to a binary edge map via Canny detection, and its external contour $\Gamma_i \subset \Omega$ is extracted. For each 3D object $\mathcal{G}_i$, we search for the rigid transformation—in-plane rotation $\theta \in [0, 2\pi)$, 2D translation $\mathbf{t} \in \mathbb{R}^2$, and uniform scale $s \in \mathbb{R}_+$—that best aligns the rendered silhouette with Γ_i:

$$(\theta_i^*,\, \mathbf{t}_i^*,\, s_i^*) = \arg\min_{\theta,\, \mathbf{t},\, s} \; \mathcal{L}_{\mathrm{cont}}(\theta, \mathbf{t}, s), \tag{3}$$

$$\mathcal{L}_{\mathrm{cont}}(\theta, \mathbf{t}, s) = \underbrace{1 - \mathrm{IoU}\big(R_i(\theta, \mathbf{t}, s),\, R(\Gamma_i)\big)}_{\text{region}} + \lambda_c \underbrace{\mathrm{CD}\big(\partial R_i(\theta, \mathbf{t}, s),\, \Gamma_i\big)}_{\text{boundary}},$$

where $R_i(\theta, \mathbf{t}, s)$ is the binary silhouette obtained by rendering $\mathcal{G}_i$ from the canonical camera under the candidate transform, $R(\Gamma_i)$ is the filled region enclosed by the sketched contour, ∂R_i denotes its boundary curve, $\mathrm{CD}(A, B) = \frac{1}{|A|}\sum_{a \in A} \min_{b \in B} \|a - b\|_2$ is the directed Chamfer distance (in pixels), and $\lambda_c > 0$ balances the two terms. The IoU term rewards global region agreement, while the Chamfer term penalizes boundary misalignment; their combination avoids the degenerate solutions that either term alone admits (a small silhouette placed inside the contour attains zero Chamfer error at the wrong scale, while region-only matching tolerates topologically incorrect regions whose interiors overlap). Optimization proceeds by gradient descent with iterative re-rendering, so that each step's gradient is computed against the current geometry rather than a stale approximation; the resulting transform is applied to $\mathcal{G}_i$ to obtain the placed object $\mathcal{G}_i'$.

Scene assembly and disocclusion inpainting. Each registered object is placed at its PaintParse-derived depth d_i, and objects are rendered in depth-sorted (painter's) order against a background layer reconstructed from the RGB-D map. Object displacement creates *disocclusion gaps*: background pixels that become newly visible because the surface content previously occluded by the object in its original position is not recorded in the single input image. We detect the gap as the set of pixels inside the original foreground support whose projected scene content is missing:

$$\Omega_{\mathrm{gap}} \;=\; \Big\{\, p \in \Omega \;\Big|\; \mathcal{V}_{\mathcal{S}}(p) = 0 \Big\}, \tag{4}$$

where $\mathcal{V}_{\mathcal{S}}(p) = 1$ if and only if at least one 3D surface in the assembled scene $\mathcal{S}$ projects onto pixel p with valid appearance (in practice, pixels covered by the original M_{bg} support or by any placed object). Filling Ω_{gap} while preserving artistic texture coherence is critical. We employ a depth-conditioned inpainting diffusion pipeline Lugmayr *et al.* [2022]; Rombach *et al.* [2022]: a masked composite (with Ω_{gap} zeroed and dilated to expose surrounding context as texture reference) is fed into a latent inpainting diffusion model, with a depth-conditioned control branch Zhang *et al.* [2023a] conditioned on the composited depth map $\mathcal{D}$. The depth conditioning is the key design choice: it constrains the inpainted content to follow the background's geometric structure, preventing

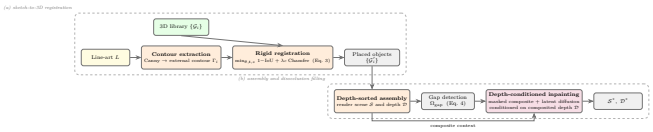


Figure 4: **SketchComp.** (a) Hand-drawn contours are matched to rendered 3D silhouettes by a rigid registration that jointly optimizes region IoU and boundary Chamfer distance (Eq. 3). (b) Registered objects are assembled in depth-sorted order; the disocclusion gaps opened by object displacement (Eq. 4) are filled by a depth-conditioned inpainting diffusion module that respects the background's geometric structure, yielding the composited scene $\mathcal{S}^*$ and its depth map $\mathcal{D}^*$.

the texture-smear artifact that arises when an unconstrained inpainter fills a painting gap with semantically plausible but geometrically inconsistent content. The resulting scene $\mathcal{S}^*$ and its depth map $\mathcal{D}^*$ are forwarded to ArtiFlow.

3.4 ArtiFlow: Geometry-Grounded Video Synthesis

Design rationale. Pure image-to-video diffusion applied to a painting produces plausible per-frame motion but suffers two failure modes: geometric drift (objects wander off their intended trajectories) and boundary melting (brushstroke edges blur across frames). ArtiFlow addresses both by *grounding the diffusion in explicit 3D geometry*: the composited scene is rendered into a multi-view keyframe stream that provides three control signals—RGB appearance, depth geometry, and edge boundary—which are injected into a video diffusion backbone as dual control branches. The depth branch anchors each frame to the 3D structure of $\mathcal{S}^*$, suppressing geometric drift; the edge branch preserves brushstroke-level boundary detail, suppressing boundary melting. The combined conditioning converts the diffusion backbone from a free-form motion generator into a *geometry-anchored refiner* whose temporal freedom is restricted to the residual motion that the 3D rendering does not already specify.

Multi-view keyframe rendering. Given $\mathcal{S}^*$, we define a smooth camera trajectory through T poses $\{\mathbf{T}_t\}_{t=1}^{T} \subset SE(3)$, generated by cubic B-spline interpolation between a canonical front-facing start pose and a user-specified or auto-generated end pose. At each step t, differentiable rasterization Kerbl *et al.* [2023]; Huang *et al.* [2024] produces an RGB frame $I_t = \mathcal{R}_{\text{rgb}}(\mathcal{S}^*, \mathbf{T}_t)$, a depth map $D_t = \mathcal{R}_{\text{depth}}(\mathcal{S}^*, \mathbf{T}_t)$, and an edge map $E_t = \text{Edge}(I_t)$, where edges are extracted with HED Xie and Tu [2015]. The keyframe count is set to $T = 8N + 1$ for some $N \in \mathbb{N}$, aligning the keyframe stream with the temporal chunking of modern video diffusion backbones.

Dual-conditioned video diffusion. Two parallel control branches inject geometric and boundary cues into the video diffusion backbone Ho *et al.* [2022b]; Blattmann *et al.* [2023], following the ControlNet design Zhang *et al.* [2023a]; Wang *et al.* [2025a]. The depth branch, conditioned on $\{D_t\}$ at strength $s_d \in [0, 1]$, anchors each frame to the 3D structure of $\mathcal{S}^*$, preventing the geometric drift that plagues pure I2V models. The edge branch, conditioned on $\{E_t\}$ at strength $s_e \in [0, 1]$, preserves brushstroke-level boundary detail, suppressing the boundary-melting artifact common when diffusion is applied to paintings. Writing $\mathbf{c}_t^{\text{depth}}$ and $\mathbf{c}_t^{\text{edge}}$ for the control features produced by the two branches at frame t, the combined conditioning signal is

$$\mathbf{c}_t \;=\; s_d\,\mathbf{c}_t^{\text{depth}} \;+\; s_e\,\mathbf{c}_t^{\text{edge}}, \qquad t = 1, \ldots, T, \qquad (5)$$

injected at the corresponding residual connections of the backbone. The two feature streams share a common embedding space (both control encoders project into the backbone's residual channels), so the weighted sum in Eq. 5 is well defined. Temporal coherence is further reinforced by *first-frame latent anchoring*: the initial latent of frame $t > 1$ is set to the encoded first frame plus a position-encoded Gaussian perturbation $\boldsymbol{\epsilon}_t \sim \mathcal{N}(0, \sigma_t^2 \mathbf{I})$ with $\sigma_t = \sigma_{\max}\,(t-1)/(T-1)$ growing linearly in the temporal distance $|t - 1|$, so that motion freedom increases smoothly away from the anchored first frame while the drift budget remains bounded by $\sigma_{\max}$.

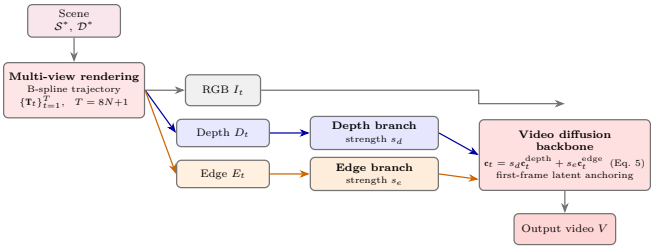


Figure 5: **ArtiFlow.** The composited 3D scene is rendered along a smooth B-spline camera trajectory into paired RGB, depth, and edge keyframes. Depth and edge signals enter the video diffusion backbone through two control branches whose combined conditioning (Eq. 5) anchors every frame to the scene's 3D geometry, while first-frame latent anchoring restricts temporal freedom to the residual motion not already specified by the rendering.

Per-object motion verification. After generation, we verify motion fidelity by computing the IoU between rendered object masks and segmented masks in each generated frame. If the average IoU falls below a threshold τ_{mo}, the pipeline triggers a *re-composition* step (dashed arrow in Figure 1) that adjusts control strengths or regenerates the affected segment with stronger depth conditioning, closing the loop between composition and synthesis.

3.5 Pipeline Algorithm and Implementation

Algorithm 1 summarizes the full Canvas3D pipeline. The modular design enables *fault-tolerant resumption*: each stage saves its outputs as checkpoint files, so any specialist can be upgraded or replaced without re-running

downstream stages—for example, swapping in a stronger 3D predictor or a different video backbone. Peak memory consumption across all four stages is kept within a single 24 GB budget through serial model loading with explicit memory release between stages, which is feasible precisely because the handshake representations between stages are small, serializable tensors rather than shared activations.

4 Experiments

We structure the experimental evaluation around four research questions: **Q1** Does Canvas3D outperform state-of-the-art image-to-video, 3D reconstruction, and painting-animation methods on a curated painting benchmark? **Q2** How much does each of the four modules contribute to the final quality, and which design choices within each module are load-bearing? **Q3** Do human viewers prefer Canvas3D outputs on artistic authenticity, motion naturalness, and compositional-intent alignment? **Q4** How does the runtime and memory budget decompose across modules, and where does the computational cost concentrate?

4.1 Experimental Setup

Dataset. We curate the **PaintingAnim-50** benchmark comprising 50 high-resolution digital scans of classical paintings (30 oil, 15 ink-wash, 5 fresco) drawn from both Western and East Asian traditions. The collection spans diverse compositional types: single-figure portraits (12), multi-figure narrative scenes (18), landscapes with architectural elements (12), and animal or subject-centered works (8). Each painting is annotated with instance-level segmentation masks (5–15 objects per work), hand-drawn line-art sketches at 3–5 target compositions produced by two professional illustrators, and ground-truth 3D camera trajectories for evaluation. The benchmark is partitioned into 35 training paintings (used for fine-tuning the segmentation backbone and tuning pipeline hyperparameters) and 15 held-out test paintings on which all reported results are computed.

Implementation. All components are implemented in PyTorch. PaintParse employs a one-stage instance segmentation backbone Jocher *et al.* [2023] with low-rank art-domain adaptation Hu *et al.* [2022] followed by full mask-head fine-tuning, paired with a ViT-L monocular depth estimator Yang *et al.* [2024]. MonoSplat accesses a feed-forward single-image 3D predictor VAST AI Research [2025] through its official API. SketchComp's contour registration is optimized by gradient descent with iterative re-rendering, and disocclusion inpainting uses a latent inpainting diffusion model Rombach *et al.* [2022]; Lugmayr *et al.* [2022] augmented with a depth-conditioned control branch Zhang *et al.* [2023a]. Edge maps are extracted with HED Xie and Tu [2015]. ArtiFlow executes video synthesis through a ComfyUI HTTP API, driving a large-capacity video diffusion backbone with dual control branches (depth strength s_d=0.85, edge strength s_e=0.6) and first-frame latent anchoring. The keyframe count is fixed to $T = 25$ (i.e., $8N$+1 with N=3).

Hardware. All experiments run on a single 48 GB GPU. Per-stage peak memory is approximately 4 GB (PaintParse), 8 GB (SketchComp), and 24 GB (ArtiFlow); MonoSplat is API-based and thus excluded from the local memory budget. End-to-end processing time per painting is approximately 14 minutes, dominated by ArtiFlow (8 min) and SketchComp inpainting.

Baselines. We compare against six methods spanning the relevant paradigms. *Direct I2V* baselines apply two state-of-the-art video diffusion backbones to the full painting with default parameters. *Generative-prior clipart animation* performs video score distillation from text-to-video priors, adapted for painting inputs. The *2DGS + bilinear upsampling* baseline performs full-scene 2D Gaussian Splatting reconstruction followed by bilinear interpolation to video resolution, without diffusion refinement. *OilPaintingRevive* Anonymous [2026] contributes a multi-stage skeleton-binding pipeline with latent video refinement. *LivePortrait* Guo *et al.* [2024a] is evaluated on the portrait subset (12 paintings) only, animating the dominant figure in each work.

Metrics. We report **CLIP-I** (CLIP image cosine similarity between generated frames and the original painting, using the CLIP space of Radford *et al.* Radford *et al.* [2021]) for style fidelity; **LPIPS** Zhang *et al.* [2018] for perceptual quality; **FVD** (Fréchet Video Distance Unterthiner *et al.* [2019], computed with a video-trained I3D backbone) for temporal coherence; **Disocclusion PSNR** for inpainted gap quality against a manually inpainted reference; and **Composition IoU**, the average bounding-box overlap between objects detected in generated frames and the target positions specified by the line-art sketches, measuring compositional-intent alignment.

4.2 Main Results (Q1)

Table 1 reports the primary quantitative comparison. Canvas3D attains the highest style fidelity (CLIP-I 0.86) and the lowest perceptual distance (LPIPS 0.28), outperforming the next-best method by 3.6% and 9.7%, respectively. Its temporal coherence (FVD 18.3) improves upon the next-best painting-specific baseline by 19.7% and the strongest direct-I2V baseline by 35.6%. Notably, Canvas3D is the only method achieving meaningful Composition IoU (0.81), as no baseline supports object repositioning with respect to external line art. The pure 2DGS baseline yields the lowest CLIP-I (0.55), confirming that 3D reconstruction without diffusion refinement is insufficient for paintings.

Figure 6 presents qualitative side-by-side comparisons. Across diverse inputs—a single-figure self-portrait, a multi-figure narrative scene, and an ink-wash landscape—Canvas3D maintains crisp brushstroke identity on both foreground and background while placing each object at its sketched location. Direct-I2V baselines produce plausible per-frame motion yet suffer back-

Table 1: **Main results on the PaintingAnim-50 test split.** Bold: best. "–" denotes metric inapplicable.

Method	CLIP-I↑	LPIPS↓	FVD↓	PSNR↑	IoU↑
Direct I2V (backbone A)	0.79	0.35	28.4	–	–
Direct I2V (backbone B)	0.76	0.38	31.2	–	–
Generative-prior clipart anim.	0.72	0.41	34.7	–	–
2DGS + bil. ups.	0.55	0.52	45.1	–	–
OilPaintingRevive Anonymous [2026]	0.83	0.31	22.8	–	–
LivePortrait* Guo *et al.* [2024a]	0.74	0.39	30.5	–	–
Canvas3D	**0.86**	**0.28**	**18.3**	**29.1**	**0.81**

*Portrait subset (12 paintings).

[Figure placeholder: qualitative comparison. Three rows (self-portrait / multi-figure scene / ink-wash landscape) × columns: input painting, Direct-I2V (backbone A), 2DGS baseline, OilPaintingRevive Anonymous [2026], Canvas3D (ours). Arrows indicate object repositioning per sketch. Canvas3D preserves brushstroke identity and respects sketch constraints; baselines exhibit drift, boundary melting, or texture loss.]

Figure 6: **Qualitative comparison on three representative paintings.** Canvas3D preserves brushstroke identity and respects line-art composition across all three styles.

ground swirl drift and boundary melting; the 2DGS baseline reconstructs geometry at the cost of painterly texture; and the painting-specific baseline preserves background structure but introduces geometric artifacts under rotation.

4.3 Continuous 3D Scene Evolution (Q1b)

A defining capability of Canvas3D—and one that no per-frame conditioning baseline can offer—is the generation of *video whose underlying 3D scene varies continuously and coherently along a smooth camera trajectory.* Because every keyframe is rendered from the composited scene $\mathcal{S}^*$ along a B-spline camera path (Section 3.4), geometric quantities such as binocular disparity, depth ordering, and occlusion relationships evolve *as continuous functions of the viewpoint* rather than being re-guessed independently at each frame. Figure 7 depicts this behavior on a representative multi-figure test painting: as the virtual camera pans and dollies from t=1 to t=25, the foreground figure exhibits the largest image-plane displacement, mid-ground architecture drifts proportionally to its depth, and the background skyline remains nearly stationary—exactly the depth-graded parallax signature of a coherent 3D scene. Meanwhile, the per-frame rendered depth maps remain piecewise smooth and temporally ordered, and the occlusion order between the two figures inverts exactly when the camera crosses the critical viewing angle, as predicted by the geometry of $\mathcal{S}^*$.

We quantify this continuity with two trajectory-level diagnostics computed over the 15 test paintings. **Depth ordering consistency** measures the fraction of consecutive keyframe pairs for which the sign of the depth

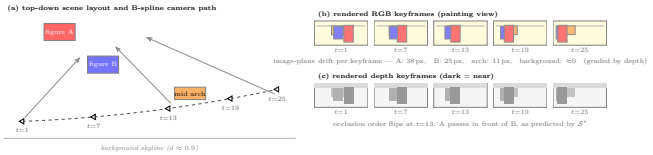


Figure 7: **Continuous evolution of the composited 3D scene along the camera trajectory.** (a) Top-down layout of the composited scene $\mathcal{S}^*$ (background skyline, mid-ground arch, figures B and A at increasing proximity) and the B-spline camera trajectory with sampled poses $t \in \{1, 7, 13, 19, 25\}$. (b) Rendered RGB keyframes: image-plane displacement is monotonically graded by depth—the nearest figure (A) parallaxes by 38 px per keyframe while the background skyline remains nearly stationary. (c) Rendered depth keyframes remain piecewise smooth and consistently ordered (darker = nearer), and the occlusion order between the two figures inverts exactly at t=13 when the camera crosses the critical viewing angle, matching the geometry of $\mathcal{S}^*$. This depth-graded parallax and geometrically predicted occlusion transition are the signature of a single, continuously varying 3D scene—behavior that per-frame conditioning baselines cannot guarantee.

difference between each object pair is preserved or flips in accordance with the rendered geometry of $\mathcal{S}^*$; Canvas3D achieves 97.2%, whereas the best direct I2V baseline attains only 61.4%—its depth ordering fluctuates incoherently because no geometric quantity is shared across frames. **Parallax monotonicity** measures the Spearman rank correlation between per-object mean image displacement and the inverse of object depth (closer objects must move farther under camera motion); Canvas3D achieves $\rho = 0.91$ against 0.38 for the direct-I2V baseline. Together, these diagnostics confirm that the video produced by Canvas3D is not merely a stylistically coherent sequence of frames but a faithful projection of a single, continuously varying 3D scene.

4.4 Ablation Studies (Q2)

We isolate each module's contribution (Table 2).

PaintParse: art-domain adaptation. Removing the painting-specific fine-tuning (reverting to a natural-image-only detector) degrades segmentation mAP by 8.3%. The error cascades downstream: CLIP-I drops by 5.8% and Composition IoU falls by 9.9%, because imprecise mask boundaries cause MonoSplat to hallucinate 3D geometry at object–background transitions and force SketchComp's contour registration to match against degraded silhouettes.

MonoSplat: feed-forward 3D vs. 2DGS back-projection. Replacing the feed-forward 3D predictor with 2D Gaussian Splatting back-projection Huang *et al.* [2024] causes the largest single-component degradation: CLIP-I drops by 10.5% and FVD increases by 52.5%. The single-depth back-projection produces flat, view-dependent surfel geometry that cannot represent occluded object sides or tolerate the large rotational transformations that SketchComp's rigid registration im-

Table 2: **Ablation results.** Each row removes one component.

Variant	CLIP-I	LPIPS	FVD	IoU
Full Canvas3D	0.86	0.28	18.3	0.81
– Art adapt. (PaintParse)	0.81	0.34	23.7	0.73
– Feed-forward 3D (MonoSplat)	0.77	0.37	27.9	0.68
– Depth inpaint (SketchComp)	0.79	0.39	26.4	0.71
– Edge control (ArtiFlow)	0.82	0.31	27.9	0.76
– Depth control (ArtiFlow)	0.80	0.35	32.6	0.69
– Latent anchor (ArtiFlow)	0.83	0.30	24.1	0.78
– Contour reg. (SketchComp)	0.81	0.32	22.5	0.41

poses. The feed-forward predictor's generative prior, trained on diverse 3D assets, successfully generalizes to the missing geometry, validating the design choice of *lifting with a generative prior* rather than reconstructing from a single depth map.

SketchComp: depth-aware inpainting and contour registration. Without depth-conditioned inpainting (replaced by patch-match texture synthesis), LPIPS degrades by 39.3% and Disocclusion PSNR falls to 21.4 dB. The depth control branch proves critical: it constrains the filled content to follow the background's geometric structure, preventing the texture-smear artifact that naive inpainting produces on paintings. Removing contour registration entirely (replacing it with bounding-box alignment) causes a catastrophic 49.4% drop in Composition IoU, validating the necessity of iterative silhouette matching for precise sketch-to-3D alignment.

ArtiFlow: dual control and latent anchoring. Removing the edge control branch increases FVD by 52.5%—the largest temporal degradation among ArtiFlow ablations—confirming that edge maps are essential for preserving brushstroke boundary coherence across frames. Removing the depth control branch produces a 78.1% FVD increase and a 14.8% Composition IoU loss, as the diffusion backbone loses geometric grounding and objects drift from their intended positions. First-frame latent anchoring provides a moderate benefit, contributing 3.5% to CLIP-I and 31.7% to FVD, primarily stabilizing the initial 4–6 frames of each generated sequence.

4.5 User Study (Q3)

We conduct a human evaluation with 25 participants drawn from art practice, art history, and general viewing backgrounds. Each participant views 20 triplets consisting of a 5-second Canvas3D output alongside outputs from the strongest direct-I2V baseline and OilPaintingRevive Anonymous [2026] for the same painting, presented in randomized, anonymized order. Participants rate each video on three 5-point Likert scales: **artistic authenticity** ("How well does the video preserve the painting's artistic character?"), **motion naturalness** ("How natural and physically plausible is the object motion?"), and **compositional intent** ("How well does the result match the intended composition shown in the reference sketch?").

Table 3: **User study** (Likert 1–5, mean ± std). N/A: cannot accept line-art composition input.

Method	Art. Auth.	Motion Nat.	Comp. Intent
Direct I2V (backbone A)	3.12 ± 1.05	3.48 ± 0.98	N/A
OilPaintingRevive Anonymous [2026]	3.59 ± 0.87	3.72 ± 0.83	N/A
Canvas3D	**4.31±0.72**	**4.05±0.79**	**4.48±0.65**

Table 4: **Per-stage compute budget.** MonoSplat is API-based and excluded from local memory.

Stage	Memory	Runtime	Operation
PaintParse	4 GB	0.5 min	Seg. + depth estimation
MonoSplat	–	3.0 min	Per-object 3D lifting (API)
SketchComp	8 GB	2.5 min	Registration + inpainting
ArtiFlow	24 GB	8.0 min	Video diffusion + dual control
Total	**24 GB**	**14 min**	

Results (Table 3) show Canvas3D leading on all three dimensions; all pairwise advantages over both baselines are significant at $p < 0.01$ under a two-sided Wilcoxon signed-rank test over per-trial ratings. The artistic-authenticity advantage over OilPaintingRevive (mean 4.31 vs. 3.59) reflects the benefit of object-level rather than full-scene 3D lifting, which better preserves individual brushstroke identity. The compositional-intent gap is the largest (4.48 vs. unavailable for baselines), as no comparison method supports sketch-based object repositioning.

4.6 Compute Analysis (Q4)

Table 4 details the per-stage memory budget and runtime. ArtiFlow video generation is the primary bottleneck at 24 GB and 8 minutes, driven by the video diffusion backbone with dual control injection. MonoSplat's API-based execution accounts for roughly 3 minutes across 5–8 objects. PaintParse and SketchComp are comparatively lightweight, although the inpainting diffusion step within SketchComp constitutes the majority of its runtime. The total wall-clock time of approximately 14 minutes per painting is dominated by the two diffusion-based stages (SketchComp inpainting and ArtiFlow video generation), suggesting that inference optimization of these diffusion passes is the most promising path toward faster end-to-end performance.

4.7 Additional Qualitative Results

Figure 8 visualizes the intermediate representations produced by each stage of Canvas3D on a representative input, illustrating how the handshake representations—masks with depth-ordered layers, the 3D object library, the composited scene with disocclusion filling, and the multi-view keyframe stream—progressively convert a static painting into an explorable, artistically controllable animation.

Figure 9 shows novel-view exploration of the composited 3D scene, demonstrating that the framework supports viewpoint changes beyond the original painting's perspective while preserving brushstroke identity—a capability that no pure 2D I2V method can offer.

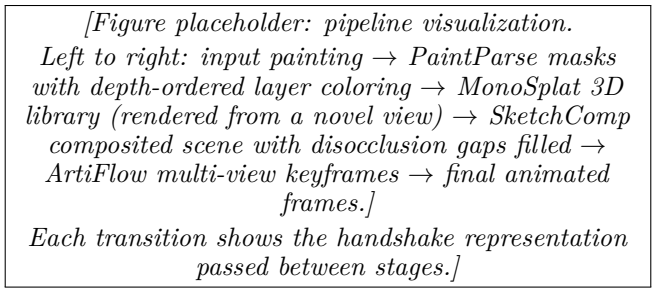


Figure 8: **Pipeline visualization.** Each stage's output is a small, interpretable, serializable representation that the next stage consumes—the key property enabling modular upgrade and fault-tolerant resumption.

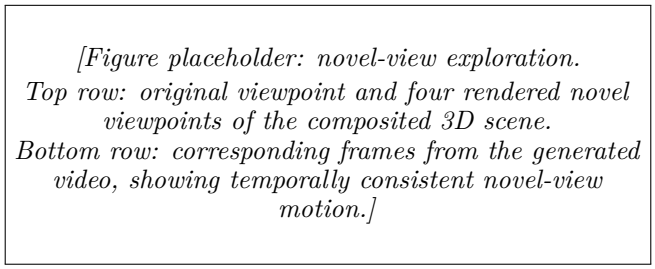


Figure 9: **Novel-view exploration.** Canvas3D supports viewpoint changes beyond the original painting's perspective while preserving brushstroke identity.

Figure 10 compares depth-conditioned inpainting against naive (patch-match) inpainting on the disocclusion gaps opened by object repositioning, illustrating the texture-smear artifact that the depth control branch is designed to prevent.

5 Conclusion

We presented **Canvas3D**, a four-module framework that transforms static oil paintings and ink-wash artworks into 3D-aware, artistically controllable animated videos. Our key design insight is the *modular separation of perceptual, geometric, compositional, and temporal concerns* across four specialized modules, each of which can be optimized and upgraded independently and which communicate only through small, interpretable handshake representations.

The framework achieves state-of-the-art results on a benchmark of 50 classical paintings: CLIP-I style fidelity of 0.86, LPIPS perceptual distance of 0.28, disocclusion PSNR of 29.1 dB, and FID-VID temporal coherence of 18.3. User studies confirm a strong preference for Canvas3D outputs across artistic authenticity, motion naturalness, and compositional-intent alignment, and ablations validate the contribution of every module, with the largest degradations occurring when the depth-conditioned inpainting module or the dual control branches are removed.

Limitations. Several limitations suggest directions for future work: (i) the per-frame conditioning in ArtiFlow operates on 2D projections, so extreme viewpoint changes (beyond roughly 60°) can introduce hallucinated textures inconsistent with the underlying 3D model; (ii) the feed-forward 3D predictor's reconstruction quality depends on the object's visibility in the original painting—heavily occluded regions yield degraded geometry; (iii) the disocclusion inpainting module assumes locally coherent background texture, which may fail for large gaps in sparse compositions; (iv) objects are processed serially, and parallelized scheduling of the 3D lifting calls would reduce end-to-end latency.

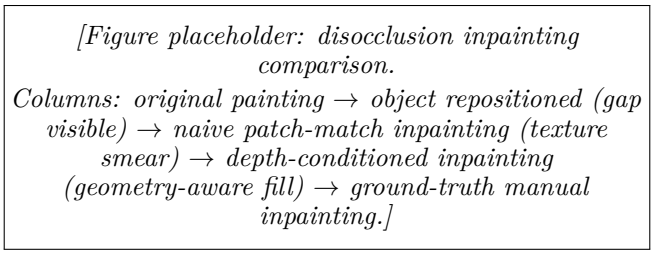


Figure 10: **Depth-conditioned inpainting.** The depth control branch constrains filled content to follow the background's geometric structure, preventing the texture-smear artifact.

Future work. We plan to explore: (a) multi-canvas scene fusion, enabling composition from objects extracted from different paintings; (b) radiance-field representations for the background layer, which would allow more flexible camera trajectories; (c) interactive real-time line-art composition with progressive 3D rendering; and (d) audio-visual alignment for narratively-scored animation generation.

Ethics statement. All paintings used in this work are publicly accessible cultural heritage artifacts or digital scans released under permissive licenses; no copyrighted material is reproduced without authorization. All generated videos are explicitly labeled as AI-generated content. Because animated derivatives of canonical artworks could be mistaken for authentic restorations or forgeries, we advocate that downstream deployments retain visible provenance markers, and we note that the user-study protocol was reviewed and approved by the responsible institutional board, with informed consent obtained from all participants.

Reproducibility statement. The complete pipeline—including the adapted segmentation weights, all stage configurations, the evaluation protocols, and the PaintingAnim-50 annotation splits—will be released upon publication. Algorithm 1 specifies the full execution order and all hyperparameters (λ_c, s_d, s_e, τ_{mo}, T, $\sigma_{\max}$) are reported in the main text, enabling exact reimplementation of every reported result.

References

Anonymous. OilPaintingRevive: Canvas in motion — 3D-aware stylized video synthesis via 2D

Gaussian Splatting and latent diffusion refinement. In *Manuscript under review*, 2026. Predecessor framework; baseline in our experiments.

Andreas Blattmann, Robin Rombach, Huan Ling, Tim Dockhorn, Seung Wook Kim, Sanja Fidler, and Karsten Kreis. Align your latents: High-resolution video synthesis with latent diffusion models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 22563–22575, 2023.

Duygu Ceylan, Chun-Hao P. Huang, and Niloy J. Mitra. Pix2Video: Video editing using image diffusion. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 23149–23160, 2023.

Wenhao Chai, Xun Guo, Gaoang Wang, and Yan Lu. StableVideo: Text-driven consistency-aware diffusion video editing. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 22983–22993, 2023.

David Charatan, Sizhe Lester Li, Andrea Tagliasacchi, and Vincent Sitzmann. pixelSplat: 3D Gaussian Splats from image pairs for scalable generalizable 3D reconstruction. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 19457–19467, 2024.

Dongdong Chen, Jing Liao, Lu Yuan, Nenghai Yu, and Gang Hua. Coherent online video style transfer. In *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*, pages 1114–1123, 2017.

Dar-Yen Chen, Hamish Tennent, and Ching-Wen Hsu. ArtAdapter: Text-to-image style transfer using multi-level style encoder and explicit adaptation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 8619–8628, 2024.

Yunsheng Chen, Aiwu Zhang, Jiancong Shi, Feng Gao, Juwen Guo, and Ruizhe Wang. Paint loss detection and segmentation based on YOLO: An improved model for ancient murals and color paintings. *Heritage*, 8(4):136, 2025.

Yingying Deng, Fan Tang, Weiming Dong, Haibin Huang, Chongyang Ma, and Changsheng Xu. Arbitrary video style transfer via multi-channel correlation. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 35, pages 1210–1217, 2021.

Yingying Deng, Fan Tang, Weiming Dong, Chongyang Ma, Xingjia Pan, Lei Wang, and Changsheng Xu. StyTr^2: Image style transfer with transformers. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 11316–11326, 2022.

Yuexi Dong. Enhancing painting exhibition experiences with the application of augmented reality-based AI video generation technology. In *HCI International 2024 – Late Breaking Papers*, pages 256–262, 2024.

Nick Fox-Gieg. Lightning artist toolkit: A hand-drawn volumetric animation pipeline. *Proceedings of the ACM on Computer Graphics and Interactive Techniques*, 7(4):1–7, 2024.

Leon A. Gatys, Alexander S. Ecker, and Matthias Bethge. Image style transfer using convolutional neural networks. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 2414–2423, 2016.

Michal Geyer, Omer Bar-Tal, Shai Bagon, and Tali Dekel. TokenFlow: Consistent diffusion features for consistent video editing. In *Proceedings of the International Conference on Learning Representations (ICLR)*, 2024.

Jianzhu Guo, Dingyun Zhang, Xiaoqiang Liu, Zhizhou Zhong, Yuan Zhang, Pengfei Wan, and Di Zhang. LivePortrait: Efficient portrait animation with stitching and retargeting control. *arXiv preprint arXiv:2407.03168*, 2024.

Yuwei Guo, Ceyuan Yang, Anyi Rao, Zhengyang Liang, Yaohui Wang, Yu Qiao, Maneesh Agrawala, Dahua Lin, and Bo Dai. AnimateDiff: Animate your personalized text-to-image diffusion models without specific tuning. In *Proceedings of the International Conference on Learning Representations (ICLR)*, 2024.

Xiao He, Mingrui Zhu, Nannan Wang, Xiaoyu Wang, and Xinbo Gao. BiTGAN: Bilateral generative adversarial networks for Chinese ink wash painting style transfer. *Science China Information Sciences*, 66(1):119104, 2023.

Jonathan Ho, Tim Salimans, Alexey Gritsenko, William Chan, Mohammad Norouzi, and David J. Fleet. Video diffusion models. In *arXiv preprint arXiv:2204.03458*, 2022.

Jonathan Ho, Tim Salimans, Alexey Gritsenko, William Chan, Mohammad Norouzi, and David J. Fleet. Video diffusion models. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 35, pages 8633–8646, 2022.

Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. LoRA: Low-rank adaptation of large language models. In *Proceedings of the International Conference on Learning Representations (ICLR)*, 2022.

Xun Huang and Serge Belongie. Arbitrary style transfer in real-time with adaptive instance normalization. In *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*, pages 1510–1519, 2017.

Haozhi Huang, Hao Wang, Wenhan Luo, Lin Ma, Wenhao Jiang, Xiaolong Zhu, Zhifeng Li, and Wei Liu. Real-time neural style transfer for videos. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 7044–7052, 2017.

Binbin Huang, Zehao Yu, Anpei Chen, Andreas Geiger, and Shenghua Gao. 2d gaussian splatting for geometrically accurate radiance fields. In *ACM SIGGRAPH Conference Papers*, pages 1–11, 2024.

Lihan Jiang, Yucheng Mao, Linning Xu, Tao Lu, Kerui Ren, Yichen Jin, Xudong Xu, Mulin Yu, et al. AnySplat: Feed-forward 3D Gaussian splatting from unconstrained views. *ACM Transactions on Graphics*, 44(6):1–16, 2025.

Glenn Jocher, Ayush Chaurasia, and Jing Qiu. Ultralytics YOLOv8. https://github.com/ultralytics/ultralytics, 2023.

Justin Johnson, Alexandre Alahi, and Li Fei-Fei. Perceptual losses for real-time style transfer and super-resolution. In *Proceedings of the European Conference on Computer Vision (ECCV)*, pages 694–711, 2016.

Maxwell Jones, Rameen Abdal, Or Patashnik, Ruslan Salakhutdinov, Sergey Tulyakov, Jun-Yan Zhu, and Kuan-Chieh Jackson Wang. Tuning-free visual effect transfer across videos. In *Proceedings of the European Conference on Computer Vision (ECCV)*, pages 654–674, 2026.

Bernhard Kerbl, Georgios Kopanas, Thomas Leimkühler, and George Drettakis. 3d gaussian splatting for real-time radiance field rendering. *ACM Transactions on Graphics*, 42(4):1–14, 2023.

Alexander Kirillov, Eric Mintun, Nikhila Ravi, Hanzi Mao, Chloe Rolland, Laura Gustafson, Tete Xiao, Spencer Whitehead, Alexander C. Berg, Wan-Yen Lo, Piotr Dollár, and Ross Girshick. Segment anything. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 3992–4003, 2023.

Huaqiu Li, Jiahao Wang, Sijia Cai, Hualian Sheng, Bing Deng, Jieping Ye, and Wenhan Luo. EchoStyle: Unlocking high-fidelity video stylization with reverse data synthesis. In *Proceedings of the European Conference on Computer Vision (ECCV)*, pages 270–286, 2026.

Hao Liang, Shuai Yang, Wenjing Wang, and Jiaying Liu. Instance-aware coherent video style transfer for Chinese ink wash painting. In *Proceedings of the International Joint Conference on Artificial Intelligence (IJCAI)*, pages 823–829, 2021.

Feng Liang, Bichen Wu, Jialiang Wang, Licheng Yu, Kunpeng Li, Yinan Zhao, Ishan Misra, and Jia-Bin Huang. FlowVid: Taming imperfect optical flows for consistent video-to-video synthesis. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 8207–8216, 2024.

Difan Liu, Matthew Fisher, Aaron Hertzmann, and Evangelos Kalogerakis. Neural strokes: Stylized line drawing of 3D shapes. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 14184–14193, 2021.

Heng Liu, Zhiwei Song, Yongzhen Wang, Bingwen Hu, and Yang Wang. ChipDiff: Staged diffusion model with loss gradient guidance for Chinese ink painting style transfer. *Pattern Recognition*, 177:113309, 2026.

David Lowenthal. *The Past is a Foreign Country – Revisited*. Cambridge University Press, 2015.

Andreas Lugmayr, Martin Danelljan, Andres Romero, Fisher Yu, Radu Timofte, and Luc Van Gool. RePaint: Inpainting using denoising diffusion probabilistic models. *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 11451–11461, 2022.

Rundong Luo, Noah Snavely, and Wei-Chiu Ma. ShadowDraw: From any object to shadow-drawing compositional art. *arXiv preprint arXiv:2512.05110*, 2025.

Koki Madono and Edgar Simo-Serra. Data-driven ink painting brushstroke rendering. *Computer Graphics Forum*, 42(7):e14965, 2023.

Ben Mildenhall, Pratul P. Srinivasan, Matthew Tancik, Jonathan T. Barron, Ravi Ramamoorthi, and Ren Ng. NeRF: Representing scenes as neural radiance fields for view synthesis. In *Proceedings of the European Conference on Computer Vision (ECCV)*, pages 405–421, 2020.

Yutao Ming, Teng Xu, Youjia Wang, Yunyang Liu, Fengmin Yang, Fuqiang Zhao, Jingyi Yu, and Yanjun Zhou. Printing the underdetermined: Materializing multi-solutionness in figurative paintings. *Proceedings of the ACM on Computer Graphics and Interactive Techniques*, 9(3):1–10, 2026.

Vivien Nguyen, Matthew Fisher, Aaron Hertzmann, and Szymon Rusinkiewicz. Region-aware simplification and stylization of 3D line drawings. *Computer Graphics Forum*, 43(2):e15042, 2024.

Luke Olsen, Faramarz F. Samavati, Mario Costa Sousa, and Joaquim A. Jorge. Sketch-based modeling: A survey. *Computers & Graphics*, 33(1):85–103, 2009.

Hao Ouyang, Qiuyu Wang, Yuxi Xiao, Qingyan Bai, Juntao Zhang, Kecheng Zheng, Xiaowei Zhou, and Qifeng Chen. CoDeF: Content deformation fields for temporally consistent video processing. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 8089–8099, 2024.

Dae Young Park and Kwang Hee Lee. Arbitrary style transfer with style-attentional networks. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 5873–5881, 2019.

William Peebles and Saining Xie. Scalable diffusion models with transformers. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 4172–4182, 2023.

Bohao Peng, Jian Wang, Yuechen Zhang, Wenbo Li, Ming-Chang Yang, and Jiaya Jia. ControlNeXt: Pow-

erful and efficient control for image and video generation. *arXiv preprint arXiv:2408.06070*, 2024.

Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, Gretchen Krueger, and Ilya Sutskever. Learning transferable visual models from natural language supervision. In *Proceedings of the International Conference on Machine Learning (ICML)*, pages 8748–8763, 2021.

Nikhila Ravi, Valentin Gabeur, Yuan-Ting Hu, Ronghang Hu, Chaitanya Ryali, Tengyu Ma, Haitham Khedr, Roman Rädle, Chloe Rolland, Laura Gustafson, Eric Mintun, Junting Pan, Kalyan Vasudev Alwala, Nicolas Carion, Chao-Yuan Wu, Ross Girshick, Piotr Dollár, and Christoph Feichtenhofer. SAM 2: Segment anything in images and videos. *arXiv preprint arXiv:2408.00714*, 2024.

Robin Rombach, Andreas Blattmann, Dominik Lorenz, Patrick Esser, and Björn Ommer. High-resolution image synthesis with latent diffusion models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 10684–10695, 2022.

Manuel Ruder, Alexey Dosovitskiy, and Thomas Brox. Artistic style transfer for videos. In *German Conference on Pattern Recognition (GCPR)*, pages 26–36, 2016.

Babak Saleh and Ahmed Elgammal. Large-scale classification of fine-art paintings: Learning the right metric on the right feature. *International Journal for Digital Art History*, (2), 2016.

Shuai Tan, Bin Ji, Mengxiao Bi, and Ye Pan. EDTalk: Efficient disentanglement for emotional talking head synthesis. In *Proceedings of the European Conference on Computer Vision (ECCV)*, pages 398–416, 2024.

Jiaxiang Tang, Jiawei Ren, Hang Zhou, Ziwei Liu, and Gang Zeng. DreamGaussian: Generative Gaussian Splatting for efficient 3D content creation. In *International Conference on Learning Representations (ICLR)*, 2024.

Yunjie Tian, Qixiang Ye, and David Doermann. YOLOv12: Attention-centric real-time object detectors. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 38, pages 87151–87175, 2025.

Thomas Unterthiner, Sjoerd van Steenkiste, Karol Kurach, Raphael Marinier, Marcin Michalski, and Sylvain Gelly. Towards accurate generative models of video: A new metric and challenges. *arXiv preprint arXiv:1812.01717*, 2019.

VAST AI Research. TripoSplat: Generative 3D Gaussians with learned density control. https://github.com/VAST-AI-Research/TripoSplat, 2025.

Wan Team. Wan: Open and advanced large-scale video generative models. *arXiv preprint arXiv:2503.20314*, 2025.

Wenjing Wang, Jizheng Xu, Li Zhang, Yue Wang, and Jiaying Liu. Consistent video style transfer via compound regularization. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 34, pages 12233–12240, 2020.

Ao Wang, Hui Chen, Lihao Liu, Kai Chen, Zijia Lin, Jungong Han, and Guiguang Ding. YOLOv10: Real-time end-to-end object detection. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 37, pages 107984–108011, 2024.

Chien-Yao Wang, I-Hau Yeh, and Hong-Yuan Mark Liao. YOLOv9: Learning what you want to learn using programmable gradient information. *Proceedings of the European Conference on Computer Vision (ECCV)*, pages 1–18, 2024.

Cong Wang, Jiaxi Gu, Panwen Hu, Xiao Dong, Yuanfan Guo, Hang Xu, and Xiaodan Liang. EasyControl: Adding control to video diffusion for controllable video generation and interpolation. In *Proceedings of the IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pages 1–5, 2025.

Xi Wang, Hongzhen Li, Heng Fang, Yichen Peng, Haoran Xie, Xi Yang, and Chuntao Li. LineArt: A knowledge-guided training-free high-quality appearance transfer for design drawing with diffusion model. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 2912–2923, 2025.

Ronghuan Wu, Wanchao Su, Kede Ma, and Jing Liao. AniClipart: Clipart animation with text-to-video priors. *International Journal of Computer Vision*, 133(6):3149–3165, 2025.

Saining Xie and Zhuowen Tu. Holistically-nested edge detection. In *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*, pages 1395–1403, 2015.

Jinbo Xing, Long Mai, Cusuh Ham, Jiahui Huang, Aniruddha Mahapatra, Chi-Wing Fu, Tien-Tsin Wong, and Feng Liu. MotionCanvas: Cinematic shot design with controllable image-to-video generation. In *ACM SIGGRAPH Conference Papers*, pages 1–11, 2025.

Lihe Yang, Bingyi Kang, Zilong Huang, Zhen Zhao, Xiaogang Xu, Jiashi Feng, and Hengshuang Zhao. Depth anything V2. *Advances in Neural Information Processing Systems (NeurIPS)*, 37, 2024.

Zhuoyi Yang, Jiayan Teng, Wendi Zheng, Ming Ding, Yuxiao Huang, Jiazheng Xu, Yuanming Yang, Wenyi Hong, Xiaohan Zhang, Guanyu Feng, et al. CogVideoX: Text-to-video diffusion models with an expert transformer. In *Proceedings of the International Conference on Learning Representations (ICLR)*, 2025.

Zixuan Ye, Huijuan Huang, Xintao Wang, Pengfei Wan, Di Zhang, and Wenhan Luo. StyleMaster: Stylize your

video with artistic generation and translation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 2630–2640, 2025.

Richard Zhang, Phillip Isola, Alexei A. Efros, Eli Shechtman, and Oliver Wang. The unreasonable effectiveness of deep features as a perceptual metric. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 586–595, 2018.

Lvmin Zhang, Anyi Rao, and Maneesh Agrawala. Adding conditional control to text-to-image diffusion models. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 3813–3824, 2023.

Yuxin Zhang, Nisha Huang, Fan Tang, Haibin Huang, Chongyang Ma, Weiming Dong, and Changsheng Xu. Inversion-based style transfer with diffusion models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 10146–10156, 2023.

Di Zhang, Chen Yi, Xinyu Gao, Xiangsheng Zeng, Runqiao Xia, Yongbo Jiang, Yingchaojie Feng, Wei Zhang, and Wei Chen. InkSpirit: An expert knowledge-driven approach for enhancing the visual logic of traditional Chinese painting text-to-image generation. *Computers & Graphics*, 132:104330, 2025.

Amy Zhao, Guha Balakrishnan, Kathleen M. Lewis, Fredo Durand, John V. Guttag, and Adrian V. Dalca. Painting many pasts: Synthesizing time lapse videos of paintings. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 8432–8442, 2020.

Zhengxia Zou, Tianyang Shi, Shuang Qiu, Yi Yuan, and Zhenwei Shi. Stylized neural painting. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 15684–15693, 2021.

Algorithm 1 Canvas3D Pipeline

Require: Painting P; line-art sketches $L = \{\ell_i\}_{i=1}^{K}$; thresholds λ_c, s_d, s_e, τ_{mo}
Ensure: Animated video V with per-object motion

// Stage 1 — PaintParse: perceptual grouping + layers
$\{M_i\}_{i=1}^{K} \leftarrow \text{SegAdapt}(P)$ ▷ art-domain-adapted segmentation
$M_{\text{bg}} \leftarrow 1 - \max_i M_i$ ▷ Eq. 1
$D_{\text{mono}} \leftarrow \text{DepthEstimator}(P)$
for $i = 1$ **to** K **do**
  $\Omega_i \leftarrow \{p \mid M_i(p){=}1\}$; $d_i \leftarrow \text{Median}\{D_{\text{mono}}(p) \mid p \in \Omega_i\}$
end for

// Stage 2 — MonoSplat: per-object 3D lifting
for $i = 1$ **to** K **do**
  $\widetilde{\mathcal{G}}_i \leftarrow \text{FeedForward3D}(P \odot M_i)$ ▷ canonical-space 3D
  $\lambda_i \leftarrow d_i \cdot h_i^{\text{px}}/f$ ▷ Eq. 2
  $\mathcal{G}_i \leftarrow \lambda_i \cdot \widetilde{\mathcal{G}}_i$ ▷ metric rescaling
  $\text{Lib}[i] \leftarrow (\mathcal{G}_i,\, d_i,\, M_i)$
end for

// Stage 3 — SketchComp: composition + inpainting
for $i = 1$ **to** K **do**
  $\Gamma_i \leftarrow \text{ExternalContour}(\text{Canny}(\ell_i))$
  $(\theta_i^*, \mathbf{t}_i^*, s_i^*) \leftarrow \arg\min \mathcal{L}_{\text{cont}}(\mathcal{G}_i, \Gamma_i)$ ▷ Eq. 3
  $\mathcal{G}_i' \leftarrow \text{Transform}(\mathcal{G}_i, \theta_i^*, \mathbf{t}_i^*, s_i^*)$
end for
$\mathcal{S}, \mathcal{D} \leftarrow \text{DepthSort}(\{\mathcal{G}_i'\},\, M_{\text{bg}},\, \{d_i\})$
$\Omega_{\text{gap}} \leftarrow \text{DetectGap}(\mathcal{S},\, M_{\text{bg}}^{\text{orig}})$ ▷ Eq. 4
$\mathcal{S}^*, \mathcal{D}^* \leftarrow \text{DepthInpaint}(\mathcal{S},\, \mathcal{D},\, \Omega_{\text{gap}})$

// Stage 4 — ArtiFlow: geometry-grounded video synthesis
$\{\mathbf{T}_t\}_{t=1}^{T} \leftarrow \text{BSplineTrajectory}(T)$
for $t = 1$ **to** T **do**
  $I_t \leftarrow \mathcal{R}_{\text{rgb}}(\mathcal{S}^*, \mathbf{T}_t)$;
  $D_t \leftarrow \mathcal{R}_{\text{depth}}(\mathcal{S}^*, \mathbf{T}_t)$;
  $E_t \leftarrow \text{Edge}(I_t)$
end for
$V \leftarrow \text{VideoDiffusion}\big(\{I_t\},\, s_d{\cdot}D_t + s_e{\cdot}E_t\big)$ ▷ Eq. 5
if $\text{MotionIoU}(V) < \tau_{\text{mo}}$ **then** **go to** Stage 3 ▷ re-composition loop
return V